



Image captioning improved visual question answering

Himanshu Sharma¹ · Anand Singh Jalal¹

Received: 9 July 2020 / Revised: 17 May 2021 / Accepted: 8 July 2021 /

Published online: 7 September 2021

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

Both Visual Question Answering (VQA) and image captioning are the problems which involve Computer Vision (CV) and Natural Language Processing (NLP) domains. In general, computer vision models are effectively utilized to represent visual contents. While NLP algorithms are used to represent the sentences. In recent years, VQA and image captioning tasks are tackled independently although they require similar type of algorithms. In this paper, a joint relationship between these two tasks is established and exploited. We present an image captioning based VQA model that uses the knowledge learnt from the image captioning task and transfers that knowledge to VQA task. We integrate the image captioning module into the VQA model by fusing the features obtained from captioning model and the attention-based visual feature. The experimental results demonstrate the improvement in the answer generation accuracy by a margin 3.45 % on VQA 1.0, 3.33% on VQA 2.0 and 1.73% on VQA-CP v2 datasets over the state-of-the-art VQA models.

Keywords Visual question answering (VQA) · Image captioning · Computer vision (CV) · Natural language processing (NLP)

1 Introduction

The extraction and understanding of the visual information in the images is still a basic problem for computer vision (CV) and natural language processing (NLP) community. Previously, the problems like image classification and object detection were focused by researchers that only deal with the recognition or detection of a narrow set of object instances or classes. Apart from these problems, the problems like image captioning [55], visual question answering [5], visual turing tests [16] and natural language based object

✉ Himanshu Sharma
himanshu.sharma@gla.ac.in

Anand Singh Jalal
asjalal@gla.ac.in

¹ Department of Computer Engineering and Applications, GLA University Mathura, Mathura, India

retrieval [26] require both visual and language understanding together with the ways of representing knowledge and reasoning potentials.

In the past few years, image captioning techniques have witnessed tremendous growth in terms of captions generated by employing deep learning methods. In past, most of the researchers have employed the sequence-to-sequence image captioning models for describing the content of an image. In such models, an image is encoded using a pre-trained CNN and represented by a fixed-length vector. Then, a natural language sentence is generated by the RNN decoder. Most of the image-captioning models are trained on the two popular datasets MSCOCO [30] and Flickr30k [39]. Depending upon the visual content and previous word, the next word of the caption is generated by the RNN decoder. Evaluation metrics used to measure the performance of these image captioning methods are BLEU [38], METEOR [12] and CIDEr [53].

Other than image captioning, the problem which uses both the domains of CV and NLP is Visual Question Answering (VQA). The aim of VQA is to generate an answer for a given image and a related question about that image in natural language. The task of VQA can be considered more difficult than other CV problems such as image classification and object detection. This is due to the fact that VQA demands better understanding and reasoning capabilities for both image and natural language questions. In traditional CV problems like image classification or object detection, answers to the questions such as image class name or finding an instance of an object given by the models is predetermined. On the other side, the user provides a question related to an image in the task of VQA. A question can be a free form open-ended, fill-in-the-blanks or a multiple options.

There are many real life applications of the VQA models. First, we can use these VQA models to answer questions asked by blind or visually impaired people. Second, we can use these models for helping young children in learning new concepts that involve concepts related to an image. Third, we can integrate a VQA system to an image-retrieval model. Fourth, we can get relevant information about an image on Internet using VQA models. Fifth, these models can summarize huge visual data for a data analyst.

Hence, it can be noticed that in VQA, we need some extra knowledge and reasoning capabilities to predict an accurate answer as compared to image captioning task. Hence VQA can be considered as an AI-complete problem [5] while image captioning provides a platform to measure how much the model understands an image. However, the evaluation of VQA models is much easier as compared to image captioning models as the answers are of few words such as “yes/no”, “color type”, “number type” or some “object name” while sentence generated by image captioning models can be long and are difficult to match with human-like captions [53].

It can be noticed that both VQA and image captioning problems link CV and NLP. The visual content of an image is effectively represented by both the tasks by employing CV approaches and NLP techniques for representing text sentences. The major difference between both of the tasks is that the image captioning task is primarily concentrate on the production of natural language explanations for a given image. On the other hand, the visual question answering task requires capturing the semantics of a given natural language question and predicts an accurate answer. Both these problems use CNNs for visual feature encoding, RNNs for modeling sentences. Still, these problems are learnt independently in previous years. Thus these two tasks are mutually correlated and thus they can be considered as the complement of each other. Thus, the focus of this paper is to investigate whether the image captioning model can improve VQA models.

To use the knowledge gained from image captioning task for the VQA task, there are two ways to use this knowledge. First approach is to develop a unified model in which

both VQA and image captioning are trained jointly. Second way is to apply the concept of transfer learning which gained popularity in most of the CV sub-problems. Consider the task of image classification, the knowledge learnt by training an image classification model on the ImageNet dataset can be straightly used for the task of object detection by the top layers tuning process. Similarly, for improving the accuracy of the VQA task, transfer learning approach can be applied to transfer the knowledge learnt from image captioning to the VQA task. So, first the model is trained on image-caption pairs. Then, this trained image captioning model is used to extract fine-grained meaningful features that makes the complement set of visual features obtained through the CNNs. Further, these fine-grained features obtained from captioning corpora and the visual features are concurrently utilized by the VQA algorithm for answer prediction. Figure 1 demonstrates an image captioning improved VQA model.

The main contributions of the proposed work are:

1. The proposed VQA model exploits the knowledge learnt from the image captioning task.
2. An image-captioning module is integrated with the VQA model by fusing the semantic visual features extracted from the image-captioning corpora with the attended visual features based on question.
3. Experiments are conducted on publically available VQA datasets and the obtained results explain that the proposed image captioning improved VQA model attains better accuracy as compared to the state-of-the-art models.

Thus, in this work, we have proposed an approach that improves the performance of a VQA model by using the captions generated for a given image. These captions are generated using the question features together with the visual features. Thus, we get image descriptions which are relevant to a given question. Experimental results show that the accuracy of proposed VQA model improves when we use these image descriptions as additional information as compared to the questions alone.

The rest paper is structured as follows: Sect. 2 discusses the related work. Section 3 mathematically defines the proposed model. Experimental results and comparative analysis are demonstrated in Sect. 4. In Sect. 5, we give conclusion and some future directions.

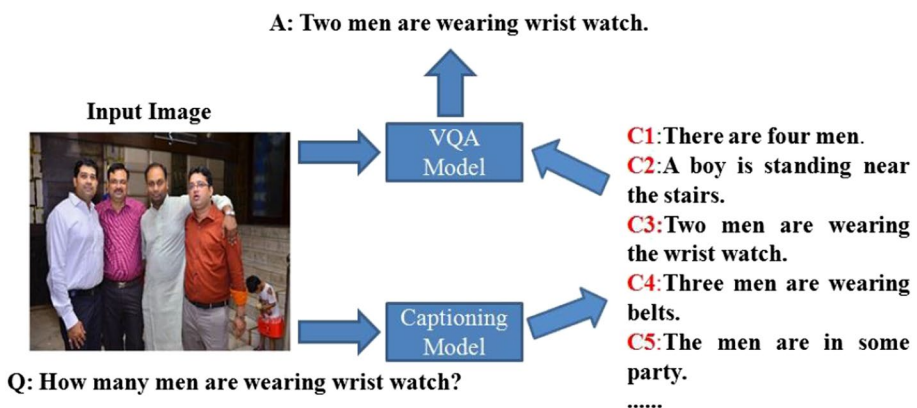


Fig. 1 An example of VQA based on image captioning

2 Related work

2.1 Image captioning

The goal of image captioning [13, 24, 29, 50, 55] is to express an image, instead of just simply recognizing an image. It includes many computer vision sub-problems such as object detection [19], action recognition [66] and image/region (fire or non-fire regions) segmentation [45]. Also, it requires knowledge of natural language processing techniques for open-ended sentence generation. These sub-problems are addressed by image captioning algorithms to different extent [24], but most image captioning models used encoder-decoder frameworks for caption generation [55].

The problem faced by many image captioning models using deep learning approach is that they can't take care of new words or objects. Reference [67] build classifiers for new class of objects by using publically accessible object recognition datasets. They used a LSTM with copy function for word-by-word generation of sentence. Novel words are selected by copy mechanism in the output caption. Reference [54] presents a novel object captioner (NOC), which uses external sources consisting of labeled images collected from datasets for object recognition and knowledge obtained using un-annotated text. Reference [60] introduces image captioning method for novel objects without producing sentences corresponding to novel objects. Yang and Xu [64] propose image captioning method based on repeatedly asking questions related to given image. Yu et al. [70] propose an image captioning model that generates captions related to a given topic. The topics are chosen from the captioning dataset. Wang et al. [58] generated captions by capturing the relationship between different regions and using attention mechanism that keeps track of earlier attention maps in addition to the current attention maps. Gupta and Jalal [18] utilized the text present in an image and combined it with the visual features to produce image captions. Yang et al. [65] add emotional concepts to describe the content of an image by gathering a candidate set of affective concepts.

Reference [42] employs a “policy network (PN)” together with “value network (VN)” for jointly producing image captions. The policy network acts as a local guide for predicting the next word by giving confidence depending upon the current state. Further, the VN provide global guidance by measuring every likely versions of the current state. Xu et al. [61] used attention mechanism together with back-propagation algorithm to describe the content of an image. Sharma and Jalal [44], Sharma et al. [46] discussed a detailed image captioning survey and use of external knowledge for image captioning.

2.2 Visual question answering

The current VQA models normally uses two types of neural networks: first type is CNN which is used for encoding images and second type is LSTM which is for encoding questions. These two encoded vectors are fused together and given to an answer decoder [5, 33, 41]. These models are improved by applying various approaches like merging of better visual and linguistic features [14, 27] and improved visual feature representation based on attention mechanism [14]. Both word-guided and question-guided attention mechanism are employed by Reference [62]. Agrawal et al. [1] explain that VQA algorithms may fail when facing the novel notions and may reach the conclusions without reading the entire question. Further, Das et al. [11] come up with the notion of

weak correlation between human attention regions and visual attended region of VQA models.

Spatial attention mechanism is added to traditional LSTM by References [22, 74]. It is added to find the weighted average of the encoded feature vectors. Reference [7] employs attention maps based on question guidance for selecting the image features corresponding to the meaning of question. Reference [63] applies stacked attention networks (SAN) for iteratively predicting the accurate answer. Goyal et al. [17] focuses on the balancing of VQA dataset by adding complementary images. Also, they focus on the role of understanding images for the task of visual question answering. Anderson et al. [3] employed both top-down and bottom-up attention modules for the task of captioning images and answering question related to an image in natural language.

Reference [49] proposes to detect the local image regions corresponding to question words by using pretrained object detection algorithm. The extracted visual features belonging to these regions are given as input to the LSTM. Zhang et al. [72] use visual relationship with attention mechanism and fuse this information to generate an answer corresponding to a given question. Liu et al. [31] explain the inverse VQA task in which they generate a question for a given image and answer pair. References [21, 35, 51] present VQA models that can read the text present in an image and answer the questions related to the text. Singh et al. [51] introduced a novel dataset names as TextVQA and Mishra et al. [35] introduced OCR-VQA-200k dataset. Hu et al. [21] extended the work of [51] by considering VQA task beyond single-step classification problem. They used pointer-augmented multi-step decoder for answer generation.

Reference [56] use DBpedia to connect visual features or concepts extracted from image to the concepts with similar semantics. Then a query is generated corresponding to the image or question features using the knowledge graph. Finally, the VQA model produces the correct answer corresponding to the query generated. Reference [57] improves [56] by using a data driven procedures for computing the relationship between images/questions features and query generated. Reference [57] employs external knowledge sources like ConceptNet and WebChild. Reference [48] presented VQA task as a multitask problem, where they integrate previous knowledge into the VQA model and train model by applying regression in vector space. Reference [23] used grid features in place of bottom-up region features for VQA task and achieves better accuracy than state-of-the art VQA and captioning models. Reference [9] presented a Counterfactual Samples Synthesizing technique for verifying the image-explainable and query sensitivity capabilities of a VQA model. Reference [59] proposed a new dataset called EST-VQA (Evidence-based Scene Text Visual Question Answering), Evidence-based Evaluation (EvE) metric. The dataset proposed by them is bilingual in nature containing both English and Chinese questions and answers sets.

Liu et al. [32] used both free form-based and detection-based attentions as the supervised attention mechanisms which are learnt from prior-knowledge of human-annotated attention maps to build a novel VQA model called adversarial learning of supervised attentions (ALSAs). Zhu et al. [75] used graph based approach to model the relationship between objects presents in an image. The relationship generated is based on the semantics of the question. Also, they proposed a difference-based graph learner which considers the difference between objects and semantics of questions to find the edges between graph nodes. Gao et al. [15] used question based object attention mechanism to generate an answer for a given question. The model uses question semantics, information related to an object and the correlation between these two feature vectors. Sharma and Jalal [47] proposes a VQA model that uses the concept of visual relationship and contextual attention

mechanism to predict an answer. The relationship between objects is generated by using graph neural network model.

3 The proposed model

Figure 2 demonstrates the block diagram of the proposed captioning-improved VQA model. First, deep CNNs, such as ResNet101, are used to encode an image using bottom-up and top-down attention model proposed by [3]. Both global and local features are extracted. We have employed GRU to encode the given input question. Further, an attention mechanism is used to obtain question dependent image features. We have used graph neural network to compute the implicit relationship between the extracted object. These image features are provided to the captioning model to generate question-based image captions. Average pooling is applied to obtain caption-based image feature representation. Further, this caption based image feature, question representation and the attended image feature are given input the LSTM [20] unit to generate the answer. These modules are discussed in detail in subsequent subsections.

3.1 Visual and question feature representation

Deep CNNs, such as ResNet101, are used to encode an image. It extracts both global and local visual features from an image using bottom-up and top-down attention model [3]. The global feature $\bar{v}_G$ are represented by the last conv-layer of the ResNet101 and has a dimension of 2048. The local features $v_1, v_2, \dots, v_L$ consist of N feature vectors. Here, each v_i corresponds to the features of a specific portion of the image. An image is evenly divided into the regions

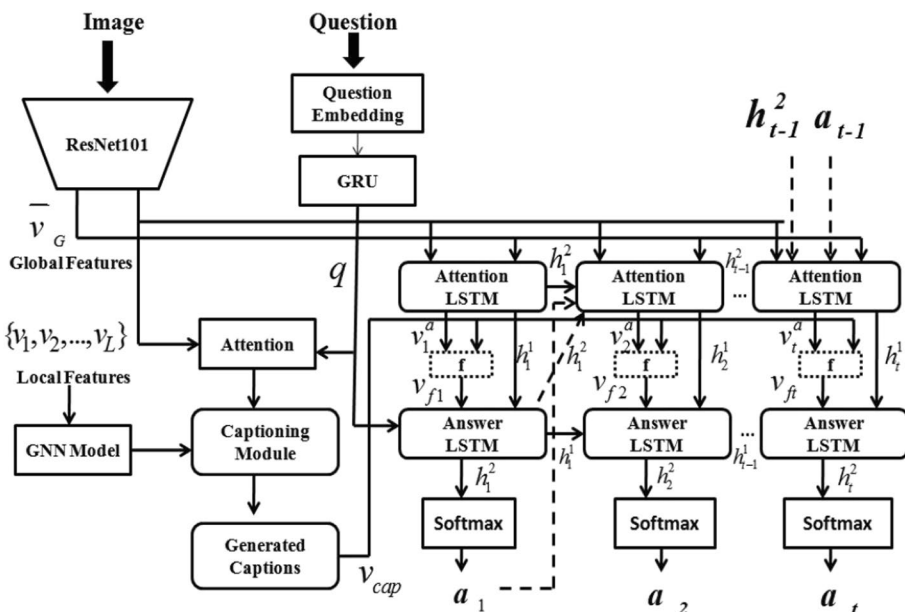


Fig. 2 Block diagram of the proposed VQA model using image captioning

corresponding to the last convolutional feature map and are defined as the local regions as in [61]. In this article, a Faster RCNN [40] is used to describe the local image segments and for extracting the local visual features through region-of-interest pooling method. We have used graph neural network to find the visual relationship representation among the different objects. The input natural language question is encoded using the conventional GRU unit having 1,280 hidden layers. The question feature are obtained as the output of the hidden state of the final word of the GRU unit and represented as q .

3.2 Attending visual features based on question

For every K image regions, we use local feature representations $\{v_1, v_2, \dots, v_L\}$ which are extracted using ResNet 101. The proposed model represents a question using the hidden state of Gated Recurrent Unit [10] (GRU). Each word of question is represented by a word embedding. Using question representation vector, the attention weights are calculated corresponding to each image regions as follows:

$$att_i = w_a^T f_a([v_i, q]) \quad (1)$$

$$\alpha_i = \text{soft max}(att_i) \quad (2)$$

where, w_a^T is the vector representing learned parameters.

The attended visual features can be represented as the summation of local visual features:

$$v'_t = \sum_{i=1}^t \alpha_i v_i \quad (3)$$

3.3 Caption generation

These question-guided visual features v'_t are fed to the LSTM-based captioning model for generating sequence of word. We have used a LSTM based image-captioning model, which utilizes a visual attention approach based on the Encoder-Decoder structure. The working of this model is explained as follow:

This model utilizes a CNN for extracting K vectors of L dimensions from the given image; where every vector corresponds to a region of the image. The proposed model also considers the fundamental convolutional layer output instead of the fully connected layer output and thus represents the image feature vector as:

$$v = \{v_1, \dots, v_L\}, v_i \in \mathbb{R}^L \quad (4)$$

In the Decoder part, LSTM is utilized for description generation. The image-dependent feature vector a for each timestamp t are also utilized to develop the context vector $z_t = \sum_{i=1}^K \alpha_{ti} v_i$. It is an incorporation of the attention approach, $\alpha_t \in \mathbb{R}^K$ is the attention weight vector of the t timestamp, which satisfies $\sum_{i=1}^K \alpha_{ti} = 1$. Here, a can be guessed by the basic neural network f_{att} and by applying softmax activation as:

$$\alpha_{ti} = \exp\{f_{att}(v_i, m_{t-1})\} \quad (5)$$

Hence, the attention based Encoder-Decoder structure can be represented as:

$$a = \text{Encoder}(I) \quad (6)$$

$$z_t = \sum_{i=1}^L \alpha_{it} v_i, \alpha_{it} \in \mathbb{R}, v_i \in \mathbb{R} \quad (7)$$

$$x_t = W_e S_t, t \in \{0, \dots, L-1\} \quad (8)$$

$$w_{t+1} = \text{Decoder}(x_t, z_t), t \in \{0, \dots, L-1\} \quad (9)$$

3.4 Image representation based on generated captions

In this stage, the hidden state vector of the LSTM when the last word has been generated corresponding to each caption is utilized for representing the caption-content. We apply average pooling over all the hidden state vectors (possibly 5 captions for an image) corresponding to each caption to generate a 512-dimensional image representation (v_{cap}) based on generated captions. We have trained the LSTM on the captions generated by humans from the training dataset. Thus, the model used for captioning focuses on image content that is of human interest for describing images.

3.5 Attention and fusion

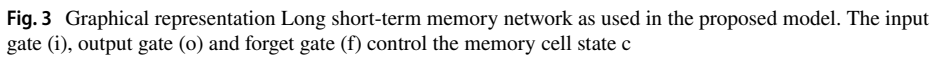
Both VQA and image captioning tasks require to focus on the target region based on the understanding of both image and question/word sequence. To be more specific, the VQA task requires generating an accurate answer based on the image and the question. In the same way, image captioning task requires to generate the next word depending upon the previously generated words and the image. Also, an image may include different objects and complicated background. Similarly, a question may contain a few words having different semantic concepts. Hence, it can be noticed that the representation of both image and question by vectors may not be able to capture the semantics of both of them. The block diagram of the LSTM unit is shown in Fig. 3. The inputs to each gate are previous hidden state (h_{t-1}), visual features ($\bar{v}_G$), current question word (q_{t-1}) and previously generated answer word (a_{t-1}).

Thus, the proposed model uses the attention module to capture the relationship between images regions and question words [15]. In the proposed work, an attention LSTM ($LSTM_{att}$) and answer generation LSTM ($LSTM_{ans}$) are employed to represent the attention between question and the image. Both of them uses the hidden states of each other for answer generation at each time step.

The inputs to the attention LSTM ($LSTM_{att}$) are previously generated answer word representation, global visual feature and previous hidden state of language LSTM.

$$h_t^1 = LSTM_{att}(a_{t-1}, h_{t-1}^1, [h_{t-1}^2; \bar{v}_G]) \quad (10)$$

where, h_t^1 is output of $LSTM_{att}$. The attended visual features can be represented as the summation of local visual features:



In this stage, the activation vector obtained from generated captions is combined with the attended visual features. The vector obtained from generated captions captures high level semantic information relative to input image feature. Thus, this vector is added to language LSTM to improve the accuracy of the answer generation. Thus, the proposed model combines both caption-based vectors with attended visual features as:

where, f is the element-wise multiplication function used for fusing both vectors.

Multiple inputs are provided to the answer LSTM such as combined visual features v_{fi} , the output from attention LSTM (h_t^1) and output vector of the answer LSTM (h_{t-1}^2) corresponding to previously generated answer word.

The proposed VQA model is able to generate multiple word answers. Thus, the VQA model is trained to maximize the probability of generated answer for the given image and question.

Mathematically, our answer generation process can be formulated as the process of word sequence generation task. Here, we used language LSTM for the generation of answer sentence word by word. The proposed model uses Softmax function over the generated answer word a_t and the output of the language LSTM as follows

$$a_t \simeq \text{softmax}(W_h h_t^2 + b_h) \quad (14)$$

For the predicted answer sentence, we can compute log-likelihood as:

$$\log \text{probability}(A \mid I, Q) = \sum_{i=1}^n \log \text{probability}(a_k / a_{1:k-1}, I, Q) \quad (15)$$

Here, $\text{probability}(a_k / a_{1:k-1}, I, Q)$ is the probability corresponding to answer word a_k , $a_{1:k-1}$ represents previously produced answer words for a given image I and question Q .

4 Experimental results

4.1 Datasets and evaluation metric

We have evaluated our proposed VQA model is evaluated on three publically available VQA 1.0 [5], VQA 2.0 [17] and VQA-CP v2 [2] datasets. VQA 1.0 dataset includes images collected from Microsoft COCO [30]. The number of questions used for training is ‘248,349’, for validation ‘121,512’ questions are employed and for testing ‘244,302’ questions are employed, that are generated for 123,287 images. Test-dev set consists of 25 % of test-standard questions. In VQA 2.0 dataset, answers to each question are balanced by minimizing the dataset priors. It contains 1.1M questions with 10 most frequent answers for each question with a total of 11.1M answers linking to MSCOCO images. The questions in both the dataset have the following types: binary (yes/no) category, number category and others category. The dataset contains 10 different free- form answers corresponding to each question, which are answered by different annotators. In VQA-CP v2 [2] (Visual Question Answering v2 under Changing Priors), new splits are created by restructuring training and validation splits of VQA v2 to change the distribution of answer per question.

In order to evaluate the quality of answers generated by the proposed mode, accuracies are computed as defined as standard VQA metric [5]. We have evaluated the results on test-std partitions. For a given answer, accuracy (Acc) is given by:

$$Acc(a_i) = \min\left(\frac{\text{count}(a_i)}{3}, 1\right) \quad (16)$$

where, $\text{count}(a_i)$: answer chosen by different annotators.

We have used Flickr30k dataset for evaluating our captioning module. Table 1 shows the overview of Flickr30k dataset.

Table 1 Statistics of Flickr30k dataset

Size	Training	Validation	Test
# Images	29000	1014	1000
# Captions	145000	5070	5000

We have evaluated our generated captions using the evaluation metric BLEU [38] and METEOR [12]. We have compared the captions generated by our captioning module with the ground truth captions to compute these evaluation metric scores.

5 Experimental setup

The proposed image captioning improved VQA model is trained by employing the AdaMax optimizer [28]. Dimensions of hidden layer of LSTM, visual features, vector representing word embedding and attended features, are all converted to 512. For each image, there are 32 object detection features in the attention model of VQA module. For encoding questions and generated captions, a popular text processing method is employed. The maximum number of words in a given question is set to 14 and then each word is converted to lower-case. A dictionary consisting of 9,500 words is built by discarding words having frequency less than 4. For training the captioning model, 5 captions for each image are used.

The training phase begins by using human-labeled captions by considering MSCOCO dataset [8]. Both VQA and image captioning modules are pre-trained for 25 epochs. Question-based captions are generated for every image-question pairs for the training, validation and testing partitions of MSCOCO dataset. We considered 5 captions for an image. The learning rate of image captioning module is set to 0.0005. It is decremented with a factor of 0.6 every 4 epochs. For avoiding over-fitting, the dropout factor is assumed to be 0.5. We consider the batch size to be 10. The scheduled sampling probability is incremented evenly after 4 epochs by a factor of 0.06. It increases until it achieves the maximum probability of 0.24. The proposed VQA model using image captioning module uses the concept of transfer learning by utilizing the knowledge acquired from image captioning model and transferring to VQA model. The proposed VQA model improved by image captioning method employs both the sum/concatenation for fusing image captioning based visual features and the attended visual features. Table 2 shows the summary of setting the hyperparameters.

5.1 Ablation study

Our image captioning improved visual question answering model is an integration of different components and needs the setting of essential learnable parameters. We have

Table 2 Hyperparameters Setting

Hyperparameters	Value
LSTM (Dimension of hidden layer)	512
Encoding dimension	512
Learning rate	0.0005
Decreasing interval for learning rate	4
Learning rate decreasing factor	0.6
Schedule sampling interval	4
Factor by which the schedule sampling is incremented	0.06
Maximum probability of the schedule sampling	0.24
Dropout Factor	0.5
Batch size	10
Epochs (Maximum Number)	25

conducted the ablation test to evaluate the contribution of each component. For each dataset, the different variations of our VQA model are trained on the train-split partition and evaluated on the valid-split partition. The proposed VQA model with different variations can be explained as:

- Baseline model: Visual features and question features are only used in this version of the proposed model. We have passed both feature vectors to the fully-connected layer.
- Baseline + Att: Attention mechanism (Att) is used together with the visual features and question features.
- Baseline + Cap: It fuses caption-based (Cap) image representation with attended visual features shown in equation 12 using sum/concatenation operations.
- Baseline + Att + Cap: It represents the full version of our image captioning improved VQA model, which utilizes visual features, question features, attention mechanism and caption based image representation. Both sum/concatenation fusion strategies are used.

Table 3 demonstrates the ablation experiments conducted to evaluate the effectiveness of each component utilized and the setting of hyperparameters considered by our VQA model. $d(V,Q)$ denotes the dimension of the vector generated by combining visual and question features, $d(Att)$ denotes the attention module dimension and $d(Cap)$ denotes the captioning module dimension. For our baseline version, $d(V,Q)=512$, $d(V,Q)=1024$ and $d(V,Q)=2048$ is tried. We get the best performance for $d(V,Q)=1024$. Further, $d(Att) = 3072$ attains the best accuracy for the Baseline + Attention version. Similarly, for Baseline + Cap version, the best performance is achieved when we tried the dimension of captioning module equal to 256. Also, the concatenation fusion strategy is used as it surpasses the addition strategy. Finally, we set $d(V,Q)=1024$, $d(Cap)=256$ and $d(Att)=3072$ in the full-version of our VQA model.

Table 3 Setting of hyperparameters and ablation experiments to evaluate the effectiveness of each module. The module or the hyperparameters used are indicated in bold

Model	Acc.	Model Size
Baseline		
$d(V,Q)=512$	56.31	15.8M
$d(V,Q)=1024$	57.12	21.8M
$d(V,Q)=2048$	56.96	47.7M
Baseline+ Att ($d(V,Q)=1024$)		
$d(Att)=1024$	60.22	24.7M
$d(Att)=2048$	61.56	27.6M
$d(Att)=3072$	62.91	30.6M
$d(Att)=4096$	62.96	33.6M
Baseline+ Cap ($d(V,Q)=1024$)		
$d(Cap)=128$	62.11	22.1M
$d(Cap)=192$	63.12	22.4M
$d(Cap)=256$	63.63	22.7M
$d(Cap)=384$	62.93	23.6M
Baseline+ Att + Cap ($d(V,Q)=1024, d(Cap)=256, d(Att)=3072$)		
Fusion using sum	66.36	33.7M
Fusion using concatenation	66.82	33.7M

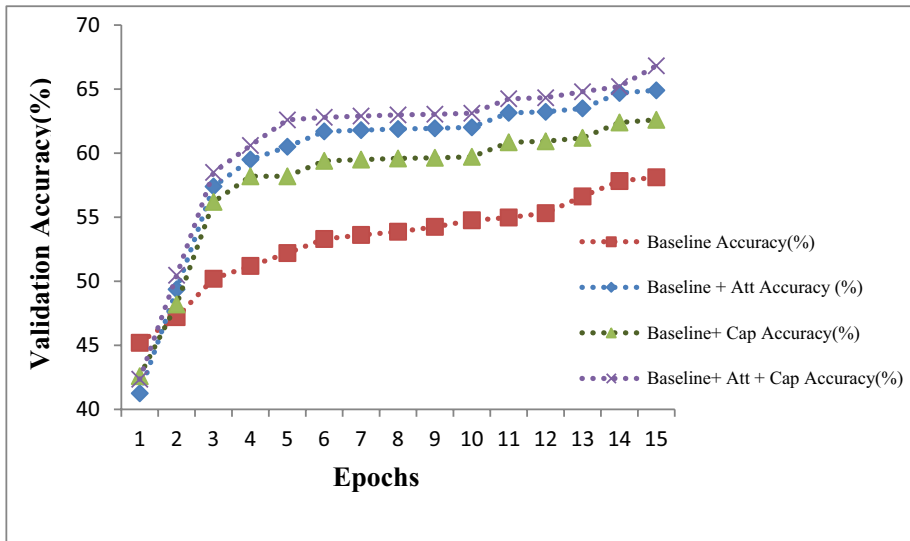


Fig. 4 Validation Accuracy vs. epochs for the proposed image captioning improved VQA model

Figure 4 demonstrates the validation accuracies of our Baseline version ($d(V,Q)=1024$), Baseline + Att version ($d(V,Q)=1024$, $d(Att)=3072$), Baseline + Cap version ($d(I,Q)=1024$, $d(Cap)=256$) and Baseline+ Att + Cap version ($d(V,Q)=1024$, $d(Att)=3072$, $d(Cap)=512$). Figure 5 shows the training losses for the Baseline version, Baseline + Att version, Baseline + Cap version and Baseline + Att + Cap version having the similar hyperparameters settings respectively.

We can summarize the overall results obtained after conducting ablation test as follows:

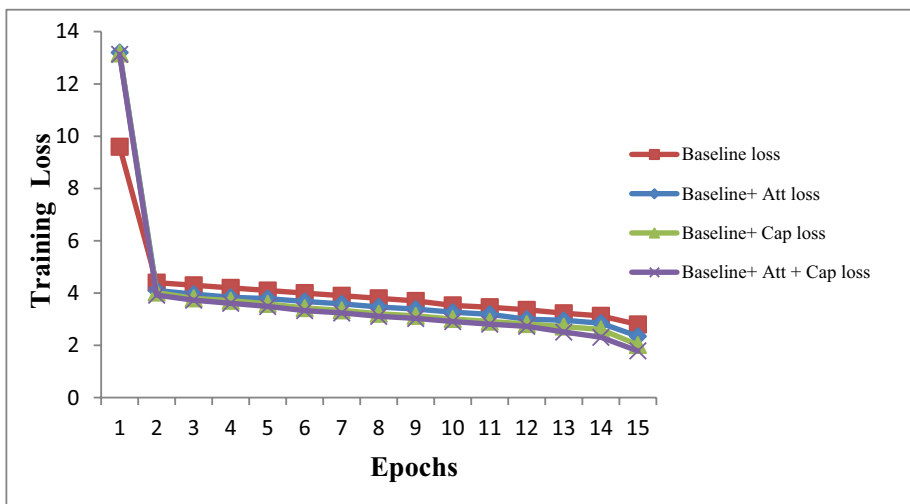


Fig. 5 Training loss vs. epochs proposed image captioning improved VQA model. For all methods cross-entropy is used

First, the use of attention mechanism with our Baseline model boosts the accuracy of the model by a margin of 5.79. It shows the importance of using attention mechanism in VQA task.

Second, when compared to the baseline model, our proposed captioning module significantly improves the accuracy from 57.12 to 63.63 with only increase of 0.9M learnable parameters. The gain in performance is attained by using the captions generated for the given image. The generated captions are able to capture the correspondence between a given image and a question, which is more fine-grained description of the visual content.

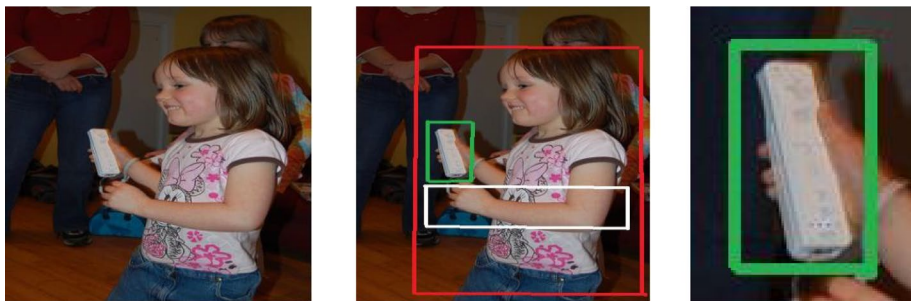
Third, the full-version of our proposed VQA model (Baseline + Att + Cap) achieves better performance as it is able to capture the correspondence between a given image and a question by generating an image caption, which is more fine-grained description of the visual content. The captioning module applies the graph neural network (GNN) which utilizes the implicit relationship among the objects present in an image. Thus, the captioning module generates fine-grained visual information and representations. The key idea behind the use of image caption for improving VQA is that a better image description must be able to capture the most salient objects or image regions which are further compared with the question sentence for the semantic correspondence.

The loss value for full-version of our proposed model decrease faster than other versions of our model, which shows that our Baseline+Att+Cap model has a better learning ability.

In Fig. 6, we have shown the generation of attention maps using the attention module. The model focuses on salient objects present in an image and their relationship based on graph neural network. As depicted in Fig. 6, visual relationship between objects is extracted by the model which is similar as attained by humans.

5.2 Results and analysis

First, we have compared our image captioning module using bottom-up and top-down attention model [3] together with the model [58] which uses graph neural network [43] for visual relationship between the salient object in an image. Our captioning module uses visual relationship modeling and attends the correct objects and captures the implicit relationship among these objects/regions to generate the human-like captions. Thus, visual relationship between objects is extracted by the model which is similar as attained by humans.



Q: What is in the hand of the girl? Caption: A girl with remote control in her hand. A: remote control

Fig. 6 Visualization of attention maps. Original image is shown in first column. Top-three attended regions are shown in second column. In third column, the answer generated by our captioning improved VQA model using attention mechanism

Table 4 Performance of captioning models on Flickr30k

Image Captioning Model	BLEU@1	BLEU@2	BLEU@3	BLEU@4	METEOR
Vinyals et al. [55]	66.3	42.3	27.7	18.3	
Karpathy and Fei-Fei [24]	57.3	36.9	24	15.7	15.3
Kiros et al. [29]	60	38	25.4	17.1	16.88
Simonyan and Zisserman [50]	60	48	28	19	-
Wang et al. [58]	69.8	51.7	37.8	27.7	21.5
Gupta and Jalal [18]	69	46	32	21.86	19.9
Xu et al. [61]	66.9	43.9	29.6	19.9	18.5
Ours (Bottom-up & Top-down + GNN)	66.8	52.1	38.8	28.1	19.1

We evaluate our captioning module on Flickr30k dataset for training. The results are demonstrated in Table 4.

Table 5, 6 and 7 depict the evaluation of our image captioning improved VQA model using the knowledge gained from image captioning module with the state-of-the-art VQA models on the VQA 1.0, VQA 2.0 and VQA-CP v2 datasets respectively.

From Table 5, the key points observed can be explained as:

First, we have compared the Baseline version of our proposed model with the Baseline + Att version. It can be noticed that when we make use of attention module, it significantly improves the performance of the baseline model. More precisely, the gain in the overall accuracy of Baseline + Att model on the test-std partition is 0.61 as compared to our Baseline model.

Second, we have compared the Baseline version of the proposed VQA model with the Baseline + Cap model. The gain in performance is attained by using the captions generated

Table 5 Performance of the proposed image captioning improved VQA model on VQA 1.0

Model	Test-standard			
	Number	Yes/No	Other	Overall
Antol et al. [5]	36.53	80.57	43.73	58.16
Kim et al. [27]	37.9	84.02	54.77	65.07
Yang et al. [63]	37.53	80.8	43.48	58.24
Zhang et al.[72]	41.33	84.18	58.58	67.33
Nguyen et al. [37]	41.27	84.61	56.83	66.66
Zhou et al.[73]	34.98	76.76	42.62	55.89
Nam et al. [36]	38.1	82.8	54	64.2
Kazemi and Elqursh [25]	39.1	82	55.2	64.6
Yu et al. [68]	40.9	83.7	53.7	64.8
Yu et al. [69]	38.9	83.8	56.3	65.8
Yu et al. [71]	39.5	85	57.4	66.9
Baseline Model	40.63	83.66	58.33	66.37
Baseline + Att Model	41.11	83.95	58.63	66.98
Baseline + Cap Model	41.31	84.16	59.32	68.21
Baseline + Att + Cap Model (concatenation)	41.63	84.76	60.12	70.35

Table 6 Performance of the proposed image captioning improved VQA model on VQA 2.0

Model	Test-standard			
	Number	Yes/No	Other	Overall
Goyal et al. [17] (Prior)	0.36	61.2	1.17	25.98
Goyal et al. [17] (Language Only)	31.55	67.01	27.37	44.26
Goyal et al. [17] (LSTM+CNN)	35.18	73.46	41.83	54.22
Goyal et al. [17] (MCB)	38.28	78.82	53.36	62.27
Fukui et al. [14]	38.28	78.82	53.36	62.27
Anderson et al. [3]	43.9	82.2	56.26	65.32
Zhang et al. [72]	44.96	83.39	58.49	67.34
Nguyen and Okatani [37]	46.93	83.89	56.9	67
Teney et al. [52]	43.9	82.2	56.26	65.67
Baseline Model	44.21	78.3	53.19	61.21
Baseline + Att Model	44.06	79.21	56.29	63.32
Baseline + Cap Model	46.98	85.31	59.18	69.91
Baseline + Att + Cap Model (concatenation)	48.23	86.63	59.98	70.67

for the given image. The generated captions are able to capture the correspondence between a given image and a question, which is more fine-grained description of the visual contents. More precisely, the gain in the overall accuracy of Baseline + Cap model on the test-std partition is 1.84 as compared to our Baseline model.

Third, the full-version of our proposed model that employs captioning module together with the attention mechanism surpasses existing VQA models. In Table 5, the full-version of our VQA model achieves overall performance of 70.35 on the test-std dataset.

Similarly, Table 6 and 7 shows the performance of the proposed VQA model on VQA 2.0 and VQA-cp v2 datasets. The proposed VQA model using captioning module together with attention mechanism outperforms the state-of-the-art models by a significant margin. In Table 6, the full-version of our proposed VQA model achieves overall accuracy of 70.67 on the test-standard partition. Also, the model achieves an accuracy of 41.27 on VQA-cp v2 dataset as shown in Table 7. We can notice that the proposed image captioning improved VQA model outperforms the other state-of-the-art VQA

Table 7 Performance of the proposed image captioning improved VQA model on VQA-CP v2

Model	Test-standard			
	Number	Yes/No	Other	Overall
Yang et al. [63]	11.14	38.35	21.74	24.96
Anderson et al. [3]	12.19	41.56	43.29	38.04
Agrawal et al. [2]	13.68	57.99	22.14	31.3
Cadene et al. [6]	13.17	42.85	45.04	39.54
Andreas et al. [4]	11.92	38.94	25.72	27.47
Malinowski et al. [34]	13.79	52.25	20.33	28.65
Baseline Model	11.12	42.31	26.32	26.33
Baseline + Att Model	11.56	47.36	29.31	29.41
Baseline + Cap Model	12.21	48.96	32.17	31.11
Baseline + Att + Cap Model (concatenation)	16.87	58.39	49.57	41.27

**Generated Captions:**

Two giraffes are standing in grassy field.
 Giraffes are standing to each other.
 The field is grassy.
 There are few trees in the field.
 One of the giraffe is eating the leaves.
Question: How many giraffes are in the field?
Answer: There are two giraffes in the field.

**Generated Captions:**

The man is holding the phone.
 The man talking on the phone..
 The man is wearing shirt and jacket.
 The man is standing under a bridge.
 The man is talking on phone under a bridge.
Question: What the man is doing?
Answer: The man is talking on phone.

Fig. 7 Results of answer prediction by the proposed image captioning improved VQA model on VQA 1.0

models by improving the accuracy by a margin of 3.45 % on VQA 1.0, 3.33% on VQA 2.0 and 1.73% on VQA-CP v2 datasets. It shows the importance of adding captions based features as the supplementary input. Other key noticing fact is that our proposed VQA model improves the performance of answer generation for ‘Number’ and ‘other’ categories of questions. This is due to the fact that captioning models are able to find the number of distinct objects present in an image which is required to answer ‘Number’ category question. Also generated captions implicitly provides additional knowledge required for the ‘Other’ category questions.

In Figs. 7 and 8, most relevant captions based on question are generated on the VQA 1.0 and VQA 2.0 datasets respectively. Also, if we change the question form, we can

**Generated Captions:**

Two persons are talking to each other.
 The person on bike is wearing a helmet.
 The window of the car is open.
 There are two cars on the road.
 There are few houses in the image.
Question: Is the person on the bike wearing the helmet?
Answer: Yes

**Generated Captions:**

The laptop is on the top of table.
 There are few chairs besides the table.
 There are tissue papers kept on the table.
 This is the image of dining room.
 The laptop is connected to a charger.
Question: Is the laptop connected to the charger?
Answer: Yes

Fig. 8 Results of answer prediction by the proposed image captioning improved VQA model on VQA 2.0



Question: Which sport is being played?

SAN [47]: Soccer

GVQA [71]: Tennis

Ours: Baseball



Question: What is the most well-known item?

SAN [47]: Carrots

GVQA [71]: Carrots

Ours: Pasta

Fig. 9 Comparison of our image captioning improved VQA model with SAN [63] and GVQA [2] models

notice that the different sets of captions are generated for the same image. Also, it can be observed that generated captions describe the objects which are crucial for answering the questions. Also the relationship between the objects present in an image is encoded by the generated captions, which also plays an important role in answering questions.

In Fig. 9, we have compared our proposed model with two deep learning based models proposed by SAN [63] and GVQA [2].

Left image: The reason for predicting incorrect answer for SAN is that SAN's attention map focuses on the ball and confused it will soccer ball. While GVQA predicts wrong answer as its attention map focuses on the field. Our model gives the correct answer as the bottom-up and top-down attention map is able to identify all the prominent objects including gloves of the player. **Right image:** Both SAN and GVQA are looking at pasta but wrongly predicting the group related to the vegetables instead of the group related to pasta.

In Fig. 10, we have shown the failure cases for the proposed model. The reason for predicting the incorrect answers is due to the incapability of the proposed model in reading the text present in images. Another reason can be the incorrect establishment of



Q. Which is the highest measurement value shown on the scale?

Proposed Model: 40

GT: 50



Q. Which bike number is on the right?

Proposed Model: 30

GT: 317

Fig. 10 Incorrect answers predicted by our proposed model

relation between text and the objects. These can be the future directions in the field of visual question answering

6 Conclusions

In this paper, we have proposed an approach that improves the performance of a VQA model by using the captions generated for a given image. These captions are generated using the question features together with the visual features. The proposed VQA model achieves better performance as it is able to capture the correspondence between a given image and a question by generating an image caption, which is more fine-grained description of the visual contents. The key idea behind the use of image caption for improving VQA is that a better image description must be able to capture the most salient objects or image regions which are further compared with the question sentence for the semantic correspondence. We evaluated the proposed VQA model on publically available VQA dataset and observed the improvement in the accuracy by a margin of 3.45 % on VQA 1.0, 3.33% on VQA 2.0 and 1.73% on VQA-CP v2 datasets. In future, we can work on developing the image captioning models by exploiting the knowledge learnt from VQA models. Another future direction is to develop VQA models that can utilize the knowledge learnt from free form Question-Answering databases.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Agrawal A, Batra D, Parikh D (2016) Analyzing the behavior of visual question answering models. arXiv preprint arXiv:1606.07356
2. Agrawal A, Batra D, Parikh D, Kembhavi A (2018) Don't just assume; look and answer: Overcoming priors for visual question answering. In Proc IEEE Conf Comput Vis Pattern Recognit 4971–4980
3. Anderson P, He X, Buehler C, Teney D, Johnson M, Gould S, Zhang L (2018) Bottom-up and top-down attention for image captioning and visual question answering. In Proc IEEE Conf Comput Vis Pattern Recognit 6077–6086
4. Andreas J, Rohrbach M, Darrell T, Klein D (2016) Neural module networks. In Proc IEEE Conf Comput Vis Pattern Recognit 39–48
5. Antol S, Agrawal A, Lu J, Mitchell M, Batra D, Zitnick CL, Parikh D (2015) Vqa: Visual question answering. In Proc IEEE Int Conf Comput Vis 2425–2433
6. Cadene R, Ben-Younes H, Cord M, Thome N (2019) Murel: Multimodal relational reasoning for visual question answering. In Proc IEEE Conf Comput Vis Pattern Recognit 1989–1998
7. Chen K, Wang J, Chen LC, Gao H, Xu W, Nevatia R (2015a) Abc-cnn: An attention based convolutional neural network for visual question answering. arXiv preprint arXiv:1511.05960
8. Chen X, Fang H, Lin TY, Vedantam R, Gupta S, Dollár P, Zitnick CL (2015b) Microsoft coco captions: Data collection and evaluation server. arXiv preprint arXiv:1504.00325
9. Chen L, Yan X, Xiao J, Zhang H, Pu S, Zhuang Y (2020) Counterfactual samples synthesizing for robust visual question answering. In Proc IEEE/CVF Conf Comput Vis Pattern Recognit 10800–10809
10. Cho K, Van Merriënboer B, Gulcehre C, Bahdanau D, Bougares F, Schwenk H, Bengio Y (2014) Learning phrase representations using RNN encoder-decoder for statistical machine translation. arXiv preprint arXiv:1406.1078

11. Das A, Agrawal H, Zitnick L, Parikh D, Batra D (2017) Human attention in visual question answering: Do humans and deep networks look at the same regions? *Comput Vis Image Underst* 163:90–100
12. Denkowski M, Lavie A (2014) Meteor universal: language specific translation evaluation for any target language. In *Proc 9th Workshop Stat Mach Transl* 376–380
13. Donahue J, Anne Hendricks L, Guadarrama S, Rohrbach M, Venugopalan S, Saenko K, Darrell T (2015) Long-term recurrent convolutional networks for visual recognition and description. In *Proceed IEEE Conf Comput Vis Pattern Recognit* 2625–2634
14. Fukui A, Park DH, Yang D, Rohrbach A, Darrell T, Rohrbach M (2016) Multimodal compact bilinear pooling for visual question answering and visual grounding. *arXiv preprint arXiv:1606.01847*
15. Gao L, Cao L, Xu X, Shao J, Song J (2020) Question-Led object attention for visual question answering. *Neurocomputing* 391:227–233
16. Geman D, Geman S, Hallonquist N, Younes L (2015) Visual turing test for computer vision systems. *Proc Natl Acad Sci* 112(12):3618–3623
17. Goyal Y, Khot T, Summers-Stay D, Batra D, Parikh D (2017) Making the V in VQA matter: Elevating the role of image understanding in Visual Question Answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 6904–6913
18. Gupta N, Jalal AS (2019) Integration of textual cues for fine-grained image captioning using deep CNN and LSTM. *Neural Comput Appl* 1–10
19. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 770–778
20. Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural computation* 9(8):1735–1780
21. Hu R, Singh A, Darrell T, Rohrbach M (2020) Iterative answer prediction with pointer-augmented multimodal transformers for textvqa. In *Proc IEEE/CVF Conf Comput Vis Pattern Recognit* 9992–10002
22. Jiang A, Wang F, Porikli F, Li Y (2015) Compositional memory for visual question answering. *arXiv preprint arXiv:1511.05676*
23. Jiang H, Misra I, Rohrbach M, Learned-Miller E, Chen X (2020) In Defense of Grid Features for Visual Question Answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* 10267–10276
24. Karpathy A, Fei-Fei L (2015) Deep visual-semantic alignments for generating image descriptions. In *Proc IEEE Conf Comput Vis Pattern Recognit* 3128–3137
25. Kazemi V, Elqursh A (2017) Show, ask, attend, and answer: A strong baseline for visual question answering. *arXiv preprint arXiv:1704.03162*
26. Kazemzadeh S, Ordonez V, Matten M, Berg T (2014). Referitgame: Referring to objects in photographs of natural scenes. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 787–798
27. Kim JH, On KW, Lim W, Kim J, Ha JW, Zhang BT (2016) Hadamard product for low-rank bilinear pooling. *arXiv preprint arXiv:1610.04325*
28. Kingma DP, Ba J (2014) Adam: A method for stochastic optimization. In *Proc 3rd Int Conf Learn Represent (ICLR)*. *arXiv preprint arXiv* 1412
29. Kiros R, Salakhutdinov R, Zemel RS (2014) Unifying visual-semantic embeddings with multimodal neural language models. *arXiv preprint arXiv:1411.2539*
30. Lin TY, Maire M, Belongie S, Hays J, Perona P, Ramanan D, ... Zitnick CL (2014) Microsoft coco: Common objects in context. In *European Conference on Computer Vision* 740–755. Springer, Cham
31. Liu F, Xiang T, Hospedales TM, Yang W, Sun C (2018) Inverse visual question answering: A new benchmark and VQA diagnosis tool. *IEEE Trans Pattern Anal Mach Intell* 42(2):460–474
32. Liu Y, Zhang X, Zhao Z, Zhang B, Cheng L, Li Z (2020) ALSA: Adversarial Learning of Supervised Attentions for Visual Question Answering. *IEEE Transactions on Cybernetics*
33. Malinowski M, Rohrbach M, Fritz M (2015) Ask your neurons: A neural-based approach to answering questions about images. In *Proceedings of the IEEE International Conference on Computer Vision* 1–9
34. Malinowski M, Doersch C, Santoro A, Battaglia P (2018) Learning visual question answering by bootstrapping hard attention. In *Proceedings of the European Conference on Computer Vision (ECCV)* 3–20
35. Mishra A, Shekhar S, Singh AK, Chakraborty A (2019) Ocr-vqa: Visual question answering by reading text in images. In *2019 International Conference on Document Analysis and Recognition (ICDAR)* 947–952. IEEE
36. Nam H, Ha JW, Kim J (2017) Dual attention networks for multimodal reasoning and matching. In *Proc IEEE Conf Comput Vis Pattern Recognit* 299–307
37. Nguyen DK, Okatani T (2018) Improved fusion of visual and language representations by dense symmetric co-attention for visual question answering. In *Proc IEEE Conf Comput Vis Pattern Recognit* 6087–6096

38. Papineni K, Roukos S, Ward T, Zhu WJ (2002) BLEU: a method for automatic evaluation of machine translation. In Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics 311–318
39. Plummer BA, Wang L, Cervantes CM, Caicedo JC, Hockenmaier J, Lazebnik S (2015) Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In Proceedings of the IEEE International Conference on Computer Vision Vis 2641–2649
40. Ren S, He K, Girshick R, Sun J (2015a) Faster R-CNN: Towards real-time object detection with region proposal networks. In NIPS
41. Ren M, Kiros R, Zemel R (2015b) Exploring models and data for image question answering. In Advances in Neural Information Processing Systems 2953–2961
42. Ren Z, Wang X, Zhang N, Lv X, Li LJ (2017) Deep reinforcement learning-based image captioning with embedding reward. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition 290–298
43. Scarselli F, Gori M, Tsoi AC, Hagenbuchner M, Monfardini G (2008) The graph neural network model. IEEE Transactions on Neural Netw 20(1):61–80
44. Sharma H, Jalal AS (2020) Incorporating external knowledge for image captioning using CNN and LSTM. Modern Phys Letters B 34(28):2050315
45. Sharma A, Singh PK, Sharma A, Kumar R (2019) An efficient architecture for the accurate detection and monitoring of an event through the sky. Comput Commun 148:115–128
46. Sharma H, Agrahari M, Singh SK, Firoj M, Mishra RK (2020) Image Captioning: A Comprehensive Survey. In 2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC) 325–328. IEEE
47. Sharma H, Jalal AS (2021) Visual question answering model based on graph neural network and contextual attention. Image Vis Comput 110:104165
48. Shevchenko V, Teney D, Dick A, Hengel AVD (2020) Visual Question Answering with Prior Class Semantics. arXiv preprint arXiv:2005.01239
49. Shih KJ, Singh S, Hoiem D (2016) Where to look: Focus regions for visual question answering. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition 4613–4621
50. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556
51. Singh A, Natarajan V, Shah M, Jiang Y, Chen X, Batra D, ... Rohrbach M (2019) Towards vqa models that can read. In Proc IEEE Conf Comput Vis Pattern Recognit 8317–8326
52. Teney D, Anderson P, He X, Van Den Hengel A (2018) Tips and tricks for visual question answering: Learnings from the 2017 challenge. In Proc IEEE Conf Comput Vis Pattern Recognit 4223–4232
53. Vedantam R, Zitnick CL, Parikh D (2015) Cider: Consensus-based image description evaluation. In Proc IEEE Conf Comput Vis Pattern Recognit 4566–4575
54. Venugopalan S, Anne Hendricks L, Rohrbach M, Mooney R, Darrell T, Saenko K (2017) Captioning images with diverse objects. In Proc IEEE Conf Comput Vis Pattern Recognit 5753–5761
55. Vinyals O, Toshev A, Bengio S, Erhan D (2015) Show and tell: A neural image caption generator. In Proc IEEE Conf Comput Vis Pattern Recognit 3156–3164
56. Wang P, Wu Q, Shen C, Hengel AVD, Dick A (2015) Explicit knowledge-based reasoning for visual question answering. arXiv preprint arXiv:1511.02570
57. Wang P, Wu Q, Shen C, Dick A, van den Hengel A (2018) Fvqa: Fact-based visual question answering. IEEE Trans Pattern Anal Mach Intell 40(10):2413–2427
58. Wang J, Wang, W, Wang, L, Wang Z, Feng DD, Tan T (2020a) Learning visual relationship and context-aware attention for image captioning. Pattern Recognit 98:107075
59. Wang X, Liu Y, Shen C, Ng CC, Luo C, Jin L, ... Wang L (2020b) On the general value of evidence, and bilingual scene-text visual question answering. In Proc IEEE/CVF Conf Comput Vis Pattern Recognit 10126–10135
60. Wu Y, Zhu L, Jiang L, Yang Y (2018) Decoupled novel object captioner. In Proc 26th ACM Int Conf Multimed 1029–1037
61. Xu K, Ba J, Kiros R, Cho K, Courville A, Salakhudinov R, ... Bengio Y (2015) Show, attend and tell: Neural image caption generation with visual attention. In International Conference on Machine Learning 2048–2057
62. Xu H, Saenko K (2016) Ask, attend and answer: Exploring question-guided spatial attention for visual question answering. In European Conference on Computer Vision 451–466 Springer, Cham
63. Yang Z, He X, Gao J, Deng L, Smola A (2016) Stacked attention networks for image question answering. In Proc IEEE Conf Comput Vis Pattern Recognit 21–29
64. Yang X, Xu C (2019). Image Captioning by Asking Questions. ACM Trans Multimed Comput Commun Appl (TOMM) 15(2s):1–19

65. Yang J, Sun Y, Liang J, Ren B, Lai SH (2019) Image captioning by incorporating affective concepts learned from both visual and textual components. *Neurocomputing* 328:56–68
66. Yao B, Fei-Fei L (2012) Recognizing human-object interactions in still images by modeling the mutual context of objects and human poses. *IEEE Trans Pattern Anal Mach Intell* 34(9):1691–1703
67. Yao T, Pan Y, Li Y, Mei T (2017) Incorporating copying mechanism in image captioning for learning novel objects. In *Proc IEEE Conf Comput Vis Pattern Recognit* 6580–6588
68. Yu D, Fu J, Mei T, Rui Y (2017a) Multi-level attention networks for visual question answering. In *Proc IEEE Conf Comput Vis Pattern Recognit* 4709–4717
69. Yu Z, Yu J, Fan J, Tao D (2017b) Multi-modal factorized bilinear pooling with co-attention learning for visual question answering. In *Proceedings of the IEEE International Conference on Computer Vision* 1821–1830
70. Yu N, Hu X, Song B, Yang J, Zhang J (2018) Topic-oriented image captioning based on order-embedding. *IEEE Trans Image Process* 28(6):2743–2754
71. Yu Z, Yu J, Xiang C, Fan J, Tao D (2018) Beyond bilinear: Generalized multimodal factorized high-order pooling for visual question answering. *IEEE Transactions on Neural Networks and Learning Systems* 29(12):5947–5959
72. Zhang W, Yu J, Hu H, Hu H, Qin Z (2020) Multimodal feature fusion by relational reasoning and attention for visual question answering. *Information Fusion* 55:116–126
73. Zhou B, Tian Y, Sukhbaatar S, Szlam A, Fergus R (2015) Simple baseline for visual question answering. *arXiv preprint arXiv:1512.02167*
74. Zhu Y, Groth O, Bernstein M, Fei-Fei L (2016) Visual7w: Grounded question answering in images. In *Proc IEEE Conf Comput Vis Pattern Recognit* 4995–5004
75. Zhu X, Mao Z, Chen Z, Li Y, Wang Z, Wang B (2020) Object-difference driven graph convolutional networks for visual question answering. *Multimed Tools Appl* 1–19

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.